

PROPERTIES OF MLE

IF YOU HAVE TWO SAMPLES HAVING THE SAME SUFFICIENT STATISTIC, THIS TWO SAMPLE WILL PROVIDE THE SAME MLE ESTIMATE

1. AGREEMENT OF MLE WITH SUFFICIENCY PRINCIPLE

2. INVARIANCE: MLE of $\tau(\theta)$, τ SOME FUNCTION,
ANY FUNCTION
"T.O."
 IS $\tau(\hat{\theta})$ (THAT IS $\hat{\tau}(\theta) = \tau(\hat{\theta})$)
ESTIMATE OF θ
"THE θ "

$\Rightarrow$ USEFUL IN PRACTICE

EX: $X \sim N(\mu, \sigma^2)$, WE KNOW $\hat{\theta} = (\bar{X}, \frac{m-1}{m} S^2)$
FOR μ FOR VARIANCE
SAMPLE VARIANCE

- IF WE WANT THE MLE OF μ^2 , THANKS TO INVARIANCE,

$$\hat{\mu}^2 = \hat{\mu}^2 = \bar{X}^2$$

SAMPLE MEAN SQUARED POPULATION STANDARD DEVIATION

- IF WE WANT THE MLE OF σ , THANKS TO INVARIANCE,

$$\hat{\sigma} = \sqrt{\hat{\sigma}^2} = \sqrt{\frac{m-1}{m}} \cdot S$$

METHODS TO EVALUATE ESTIMATORS

COMPARE THE DIFFERENT ESTIMATORS TO SEE HOW WELL THEY CAN DO THE ESTIMATIONS

WE COMPARE ESTIMATORS OF θ , IN TERMS OF MSE

(MEAN SQUARED ERROR)

$$MSE_{\theta}(T) = E[(T - \theta)^2]$$

WE WANT TO MEASURE HOW WRONG IS THE ESTIMATOR. AKA HOW DISTANT IS THE ESTIMATOR FROM THE TRUE PARAMETER

PARAMETER WE WANT TO ESTIMATE

ESTIMATOR

MEAN SQUARED ERROR

• AN ESTIMATOR T^* OF θ IS MORE EFFICIENT THAN T IF

LOWER MSE $\Rightarrow$ BETTER ESTIMATOR

$$MSE_{\theta}(T^*) \leq MSE_{\theta}(T) \quad \forall \theta \in \Theta$$

SMALLER ERROR

A STRICT INEQUALITY FOR AT LEAST ONE θ

• ALTERNATIVE MEASURES: $E[|T - \theta|]$ (MEAN ABSOLUTE ERROR)

BUT LESS USED SINCE MSE IS EASIER TO TREAT

MSE CAN BE DECOMPOSED AS FOLLOWS:

$$\begin{aligned}
 \text{MSE}_\theta(T) &= E[(T - \theta)^2] = E\left[\left(\overbrace{(T - E(T))}^{\uparrow 0} + \underbrace{(E(T) - \theta)}_{=0}\right)^2\right] \\
 &= E[(T - E(T))^2] + E[(E(T) - \theta)^2] + 2(E(T) - \theta)E(T - E(T)) \\
 &= \underbrace{V(T)}_{\substack{\text{VARIABILITY} \\ \text{OF ESTIMATOR}}} + \underbrace{(E(T) - \theta)^2}_{\substack{\text{DISTORTION OF ESTIMATOR} \\ \text{(BIAS)}}}
 \end{aligned}$$

THEN $\boxed{\text{MSE}_\theta(T) = V(T) + B_\theta(T)^2}$ WHERE

$$B_\theta(T) = E(T) - \theta \begin{cases} > 0 & \text{OVERESTIMATION (POSITIVE BIAS)} \\ = 0 & \text{UNBIASED T (NULL BIAS)} \\ < 0 & \text{UNDERESTIMATION (NEGATIVE BIAS)} \end{cases}$$

TYPICALLY YOU DO NOT HAVE AN ESTIMATION WHICH IS MORE EFFICIENT FOR ANY POSSIBLE θ .

EX: $X \sim N(\mu, \sigma^2)$. WE HAVE 2 ESTIMATORS FOR σ^2 (POP. VARIANCE):

TO COMPUTE MSE WE NEED:

- VARIANCE ESTIMATOR
- EXPECTED VALUE OF THE ESTIMATOR

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \text{AND} \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

COMPARE THE TWO ESTIMATORS TO FIND WHICH IS BETTER IN TERMS OF MSE

1° FOR S^2 : $E(S^2) = \sigma^2$ AND $V(S^2) = \frac{2}{n-1} \sigma^4$

2° FOR $\hat{\sigma}^2$: $E(\hat{\sigma}^2) = E\left(\frac{n-1}{n} S^2\right) = \frac{n-1}{n} E(S^2) = \frac{n-1}{n} \sigma^2 < \sigma^2$

$$\begin{aligned}
 V(\hat{\sigma}^2) &= V\left(\frac{n-1}{n} S^2\right) = \frac{(n-1)^2}{n^2} V(S^2) = \frac{(n-1)^2}{n^2} \cdot \frac{2}{n-1} \sigma^4 \\
 &= \frac{2(n-1)}{n^2} \sigma^4
 \end{aligned}$$

THEN $\text{MSE}_{\sigma^2}(S^2) = V(S^2) + B_{\sigma^2}(S^2)^2 = \frac{2}{n-1} \sigma^4 + \underbrace{(E(S^2) - \sigma^2)^2}_{=0}$

$$= \frac{2}{n-1} \sigma^4$$

S^2 UNBIASED ESTIMATOR OF σ^2

$$\begin{aligned}
 \text{MSE}_{\hat{\sigma}^2}(\hat{\sigma}^2) &= V(\hat{\sigma}^2) + B_{\hat{\sigma}^2}(\hat{\sigma}^2)^2 \\
 &= \frac{2(m-1)}{m^2} \sigma^4 + \left(\frac{m-1}{m} \sigma^2 - \sigma^2 \right)^2 \\
 &= \frac{2(m-1)}{m^2} \sigma^4 + \frac{1}{m^2} \sigma^4 = \frac{2m-1}{m^2} \sigma^4
 \end{aligned}$$

NOT IN RATIO THE TWO MSE

$$\frac{\text{MSE}_{\hat{\sigma}^2}(\hat{\sigma}^2)}{\text{MSE}_{S^2}(S^2)} = \frac{\frac{2m-1}{m^2} \sigma^4}{\frac{m-1}{2} \sigma^4} = \left(\frac{m-1}{m} \right) \left(\frac{2m-1}{2m} \right) = \underbrace{\left(1 - \frac{1}{m} \right)}_{< 1} \underbrace{\left(1 - \frac{1}{2m} \right)}_{< 1} < 1$$

$$\Rightarrow \text{MSE}_{\hat{\sigma}^2}(\hat{\sigma}^2) < \text{MSE}_{S^2}(S^2)$$

EVEN IF S^2 IS UNBIASED, WE PREFER $\hat{\sigma}^2$ BECAUSE OF ITS LOWER MSE

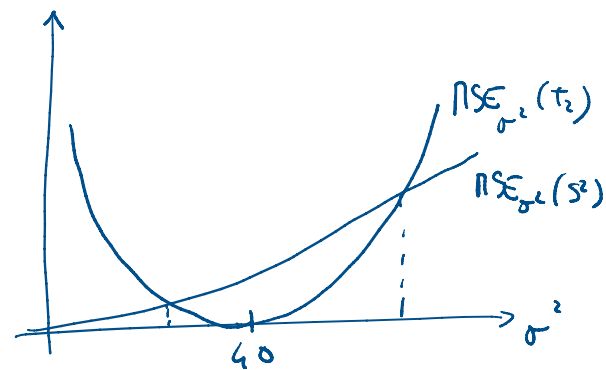
EX: $X \sim N(\mu, \sigma^2)$, 2 ESTIMATORS FOR σ^2 : $T_1 = S^2$, $T_2 = 40$

$$\text{MSE}_{\hat{\sigma}^2}(T_2) = \frac{2}{m-1} \sigma^4$$

$$\text{MSE}_{\hat{\sigma}^2}(T_2) = V(T_2) + B_{\hat{\sigma}^2}(T_2)^2 = 0 + (40 - \sigma^2)^2 = (\sigma^2 - 40)^2$$

WHEN $\sigma^2 = 40$, T_2 IS BETTER!

$\Rightarrow$ AN ESTIMATION WITH UNIFORMLY MINIMUM MSE DOES NOT EXIST



REGION AROUND 40 AT WHICH T_2 HAS LOWER MSE

TO BE ABLE TO FIND AN ESTIMATOR THAT IS BETTER THAN ANY OTHER ESTIMATOR WE HAVE TO RESTRICT THE POSSIBLE ESTIMATORS WE ARE COMPARING $\Rightarrow$ RESTRICT OURSELVES TO UNBIASED ESTIMATORS

CLASS OF ESTIMATORS IS TOO LARGE TO FIND A UNIFORMLY BETTER T
 $\Rightarrow$ TO BE ABLE TO FIND IT, WE RESTRICT TO UNBIASED ESTIMATORS

$\Rightarrow$ AN UNBIASED ESTIMATOR T IS SUCH THAT $E(T) = \theta \quad \forall \theta \in (\Theta)$

$\Rightarrow$ "2ND PIECE OF MSE = 0" $\Rightarrow$ $MSE_{\theta}(T) = V(T)$ WHEN T UNBIASED

$\Rightarrow T$ WITH MIN MSE IS T WITH MIN VARIANCE

$\rightarrow$ (UMVUE^{ESTIMATOR}: UNIFORM MINIMUM VARIANCE UNBIASED ESTIMATOR)
AMONG ALL ESTIMATORS WHICH ARE UNBIASED

$V(T^*) \leq V(T) \quad \forall \theta \in (\Theta)$ WHEN T^* IS UMVUE,
FOR T ANY OTHER UNBIASED ESTIMATOR

EX: $X \sim \text{any}$, $\bar{X}$ UNBIASED FOR $\mu \Rightarrow MSE_{\mu}(\bar{X}) = V(\bar{X}) = \sigma^2/n$

ALSO: S^2 UNBIASED FOR σ^2

! THE MLE CAN PRODUCE BIAS ESTIMATOR

ON THE OTHER HAND: $\hat{\sigma}^2$ IS BIASED FOR σ^2

$$B_{\sigma^2}(\hat{\sigma}^2) = \frac{n-1}{n} \sigma^2 - \sigma^2 = -\frac{1}{n} \sigma^2 \quad \text{NEGATIVE BIAS}$$

EX: $X \sim \text{Poisson}(\lambda) \quad E(X) = V(X) = \lambda$

THEN BOTH $\bar{X}$ AND S^2 ARE UNBIASED ESTIMATIONS OF λ

WHICH ONE DO WE PREFER? ... ANSWER LATER ...

CRAMER-RAO INEQUALITY

• THE BEST MSE AN UNBIASED ESTIMATOR CAN HAVE
• AN UNBIASED ESTIMATOR ABOVE A CERTAIN LOWER BOUND GIVEN BY THE CRAMER-RAO INEQUALITY

IT GIVES A LOWER BOUND FOR THE VARIANCE (THE MSE) OF
ANY UNBIASED ESTIMATOR

UNDER REGULARITY CONDITIONS,

ANY ESTIMATOR OF θ THAT IS UNBIASED

$$V(T) \geq \frac{1}{E\left[\left(\frac{\partial}{\partial \theta} \ln f(X; \theta)\right)^2\right]}$$

WHERE T IS
ANY UNBIASED
ESTIMATOR OF θ

REFORMULATED (UNDER ADDITIONAL REGULARITY CONDITIONS)

AS:

$$V(T) \geq \frac{1}{I(\theta)}$$

WHERE

$$I(\theta) = E \left[- \frac{\partial^2}{\partial \theta^2} \ln f(x; \theta) \right]$$

IS THE FISHER INFORMATION

FURTHERMORE: WE CAN SAY THAT $T=t(x)$ ATTAINS THE LOWER BOUND

$$1/I(\theta) \text{ IFF } \frac{\partial}{\partial \theta} \ln f(x; \theta) = a(\theta) [t(x) - \theta]$$

→ SOME FUNCTION OF θ

EX: $X \sim \text{Pois}(\lambda)$. $f(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$, $\ln f(x; \lambda) = x \cdot \ln \lambda - \lambda - \ln x!$,

$$\ell(\lambda; x) = \sum_{i=1}^n \ln f(x_i; \lambda) = \ln \lambda \cdot \sum_{i=1}^n x_i - n\lambda - \sum_{i=1}^n \ln x_i!$$

$$\frac{\partial}{\partial \lambda} \ell(\lambda; x) = \frac{\sum_{i=1}^n x_i}{\lambda} - n \Rightarrow I(\lambda) = E \left[- \frac{\partial^2}{\partial \lambda^2} \ln f(x; \lambda) \right]$$

$$\frac{\partial^2}{\partial \lambda^2} \ell(\lambda; x) = - \frac{\sum_{i=1}^n x_i}{\lambda^2} \Rightarrow = E \left[\frac{\sum_{i=1}^n x_i}{\lambda^2} \right]$$

$$= \frac{1}{\lambda^2} \sum_{i=1}^n E(x_i) = \frac{1}{\lambda^2} \cdot \lambda \sum_{i=1}^n 1 = \frac{n}{\lambda}$$

LOWER BOUND

ESTIMATOR

CRAMER-RAO LB : FOR ANY UNBIASED T , $V(T) \geq \frac{1}{I(\lambda)} = \frac{\lambda}{n}$

$$\Rightarrow V(\bar{x}) = \frac{\text{POP. VARIANCE}}{n} = \frac{\lambda}{n} \Rightarrow \bar{x} \text{ IS UMVUE.}$$

THIS IS ALSO CONFIRMED BY:

$$\frac{\partial}{\partial \lambda} \ell(\lambda; x) = \frac{\sum_{i=1}^n x_i}{\lambda} - n = \frac{n \bar{x}}{\lambda} - n = \underbrace{\left[\frac{n}{\lambda} \right]}_{a(\theta)} (\underbrace{\bar{x}}_{t(x)} - \lambda)$$

Ex: $X \sim \text{Exp}(\lambda)$. $f(x; \lambda) = \frac{1}{\lambda} e^{-x/\lambda}$, $x > 0$, $E(X) = \lambda$, $V(X) = \lambda^2$

$$\ln f(x; \lambda) = -\frac{x}{\lambda} - \ln \lambda$$

$$\ell(\lambda; \underline{x}) = -n \ln \lambda - \frac{1}{\lambda} \sum_i x_i$$

$$\frac{\partial}{\partial \lambda} \ell(\lambda; \underline{x}) = -\frac{n}{\lambda} + \frac{1}{\lambda^2} \sum_i x_i$$

$$\frac{\partial^2}{\partial \lambda^2} \ell(\lambda; \underline{x}) = \frac{n}{\lambda^2} - \frac{2}{\lambda^3} \sum_i x_i$$

$$\Rightarrow I(\lambda) = E \left[-\frac{\partial^2}{\partial \lambda^2} \ell(\lambda; \underline{x}) \right] = E \left[\frac{2}{\lambda^3} \sum_i x_i - \frac{n}{\lambda^2} \right]$$

$$= \frac{2}{\lambda^3} \sum_{i=1}^n E(x_i) - \frac{n}{\lambda^2} = \frac{2n}{\lambda^2} - \frac{n}{\lambda^2} = \frac{n}{\lambda^2}$$

$\Rightarrow$ From C-R LB: For any unbiased T , $V(T) \geq \lambda^2/n$

$$\frac{\partial}{\partial \lambda} \ell(\lambda; \underline{x}) = -\frac{n}{\lambda} + \frac{1}{\lambda^2} \sum_i x_i = -\frac{n}{\lambda} + \frac{n \bar{x}}{\lambda^2} = \left[\frac{n}{\lambda^2} \right] (\bar{x} - \lambda)$$

$a(\theta) \rightarrow \text{DEPENDS ON } \lambda$

$\Rightarrow \bar{x}$ IS UNIVUE AND REACHES THE CR-LB . INDEED:

$$V(\bar{x}) = \frac{\text{Pop. VARIANCE}}{n} = \lambda^2/n = 1/I(\theta) \quad \therefore$$